

Introduction To Pathology

The Discipline Of Pathology

GOAL

- ▶ The principal goal: understanding of the mechanisms of disease(pathogenesis) as a foundation for dealing with a vast amount of clinical information the student will encounter in later clinical years
- ▶ Core aspects of a disease process studied in pathology : etiology , pathogenesis , morphologic changes , clinical manifestation and prognosis

The Principal Objectives

- ▶ To use pathology to facilitate medical education
- ▶ Understanding mechanisms is more a function of logic than of a memory
- ▶ To leave students with a lasting knowledge of pathology
- ▶ To use pathology as the scientific basis of the "art" of medicine

What Is Pathology

- ▶ Pathology literally is the study (logos) of suffering (pathos)
- ▶ Pathology is a **bridging discipline** **specifically** devoted to the **study of the structural , biochemical and functional** changes in cells, tissues & organs **that underlie diseases**
- ▶ **Structural and functional changes → Specific for a disease → Make diagnosis**
- ▶ Pathology is the scientific study of disease
- ▶ Pathology is the foundation of medical science & practice

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- ▶ Pathology attempts to **explore the “whys” and “wherefores” of the signs and symptoms** of diseases
 - ▶ Provides basis for therapy and care → bridges basic sciences and clinical medicine.

- ▶ Four aspects of a disease process that form core of pathology
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1. **Etiology** : cause of a disease

- ▶ Two major classes of etiologic factors
- ▶ Genetic – inherited mutations
- ▶ Acquired – infectious , nutritional, chemical , physical

2. **Pathogenesis** : sequence of events in the response of cells /tissues to the etiologic agent , from the initial stimulus to ultimate expression of the disease.

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- ▶ 3. **Morphologic / molecular changes** – structural alterations in cells / tissues => characteristic of a disease or diagnostic of an etiology
 - ▶ 4. **Functional derangements and clinical manifestations** : morphologic / molecular changes result in functional abnormalities => clinical manifestations (symptoms and signs) and progress (clinical course and outcome)

The scope of Pathology

- ▶ Scientific knowledge about **human diseases** is derived from observations on patients or , by analogy, from experimental studies on animals & cell cultures.
- ▶ **Clinical medicine** is based on a longitudinal approach to a patient's illness
- ▶ **Clinical pathology** is more concerned with a cross-sectional analysis at the level of the disease itself, studied in depth- the cause & mechanisms of the disease, & the effects of the disease upon the various organs & systems

General & Systematic Pathology

- ▶ **General Pathology** : reaction of cells and tissues to abnormal stimuli and inherited defects.
- ▶ **General Pathology**: is our current understanding of the causation , mechanisms , and characteristics of the major categories of disease (e.g. congenital versus acquired diseases, inflammation, tumors, degenerations, ...)
- ▶ **General Pathology** is the foundation of knowledge that has to be laid down

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- ▶ **Systematic Pathology** : is our current knowledge of specific diseases as they affect individual organs or systems(e.g. appendicitis, lung cancer, atheroma).
 - ▶ Alterations in specialized organs and tissues
 - ▶ “Systematic” should not be confused with “Systemic”

Subdivisions of Pathology

- ▶ **Histopathology**: the investigation & diagnosis of disease from the examination of tissues
- ▶ **Cytopathology**: the investigation & the diagnosis of disease from the examination of isolated cells
- ▶ **Haematology**: the study of the disorders of the cellular & coagulable components of blood

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- ▶ **Toxicology**: the study of the effects of known or suspected poisons
 - ▶ **Forensic pathology**: the application of pathology to legal purposes(e.g. investigation of death in suspicious circumstances)
 - ▶ **Chemical pathology** : the study & diagnosis of disease from the chemical changes in tissues & fluids

Techniques Of pathology

(Diagnostic techniques used in Pathology)

- ▶ **Light Microscopy**- the structure of tissues & cells in health & disease
- ▶ **Histochemistry**- is the study of the chemistry of tissues
- ▶ **Immunohistochemistry**- employ antibodies(immunoglobulins with antigen specificity) to visualize substances in tissues sections or cell preparations

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- ▶ **Electron Microscopy-** study of disorders at an organelle level, & to the demonstration of viruses in tissue samples
 - ▶ **Biochemical techniques-** fluid & electrolyte homeostasis, serum enzyme assays e.g.raised levels of cardiac enzymes in the blood
 - ▶ **Haematological techniques-** in the diagnosis & study of blood disorders

► Molecular Pathology-

- * many advances come from the relatively new science of molecular pathology
- * **defects in the chemical structure of molecules arising from errors in the genome, using IN SITU HYBRIDISATION**

e.g. Hg molecule, collagen molecule, alterations in the genome governing the control of cell & tissue growth

- ▶ Microbiologic techniques – identify specific organisms

AFS – to detect M. tuberculosis , PAS – fungal elements

- ▶ Histopathology

- ▶ Cytopathology

- ▶ Autopsy

AUTOPSIES

- ▶ Autopsy (necropsy , post- mortem examination)
- ▶ Autopsy means to ‘see for oneself’
- ▶ Autopsies are used for:
 - determining the cause of death
 - audit of the accuracy of clinical diagnosis
 - education of undergraduates & postgraduates

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- ▶ Research into the causes and mechanisms of disease
 - ▶ Gathering accurate statistics about disease incidence
 - ▶ Investigation of new disease entities
 - ▶ There has been a regrettable decline in the autopsy rate during the latter half of the 20th century

Learning Pathology

- ▶ There are two apparent difficulties that face the new student of pathology: LANGUAGE (terms / medical words) & PROCESS
- ▶ The student must not confuse the learning of the language with the learning of the mechanisms of disease and their effects on individual organs and patients
e.g. the term hyperplasia

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- ▶ **Pathology** is learnt through a variety of media
 - text book
 - relatively didactic lectures
 - tutorials
 - demonstrations(gross & microscopic examination of diseased tissues)
 - post – mortem teaching
 - problem- solving orientated practical classes
 - Reading reference books

- ▶ Disease Mechanisms constitute general pathology
- ▶ A logical & orderly way of thinking about diseases & their characteristics must be cultivated
 - definition
 - incidence
 - etiology
 - pathogenesis
 - Morphologic changes and functional disturbances (pathological and clinical features)
 - complications and sequelae
 - prognosis

Making Diagnosis

- ▶ **Diagnosis** is the act of naming a disease in an individual patient

- ▶ The process of making diagnoses involves:
 - * taking a clinical history to document symptoms,
 - * examining the patient for clinical signs
- ↓ ↓
- provisional diagnosis
- * if necessary, performing investigations guided by the provisional diagnosis based on signs & symptoms

- * What is the role of pathologists in this process??

Diagnostic Pathology

- ▶ Identify nature and progression of disease by studying morphologic changes in tissues and chemical alteration in patients.
- ▶ In living patients we investigate & diagnose their illness by applying pathological methods to the examination of **TISSUE BIOPSIES** and **collected cells**

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- ▶ **Biopsies** - are samples of tissue removed from a patient for diagnostic purposes.
 - ▶ Two types
 - ▶ **Excisional** – taking the whole of diseased structure
 - ▶ **Incisional** – part of the diseased structure taken

 - ▶ **Resections specimens** - are the whole or part of an organ removed for a previously diagnosed condition.

- ▶ **Biopsies** - can be obtained by a variety of methods: needle biopsy, endoscopic biopsy ,incisional biopsy...
- ▶ **Cytopathology** - involves the examination and interpretation of dispersed cells rather than solid tissues, usually for the diagnosis of cancer & pre-cancerous lesions and inflammatory lesions.
- ▶ These cells can be obtained by a variety of methods according to the organ being investigated

- ▶ Exfoliative cytology: cells shed spontaneously from an epithelial surface / serosal surface

Eg .Fluid cytology :cells withdrawn from the fluid in which they are suspended

. Cells from sputum sample

- ▶ Abrasive cytology — remove / dislodge cells from epithelial surface by an instrument

Eg .Washings: cells flushed out of an organ using an irrigating fluid

. Pap test

- ▶ Fine –needle aspiration cytology(FNAC / FNA) : cells sucked out of a solid tissue using a thin needle attached to a syringe.

FNA => advantages

- ▶ **Usually first line of pathologic investigation**
- ▶ **SAFE**
- ▶ **It is Simple, Accurate, Fast, and Economic, as well as *safe***
- ▶ ***Deep organs not easily accessible by surgical incision like lungs, mediastinum***